

Vishal Jeyam

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EDUCATION

Master of Science in Computer Science | University of Arkansas, Fayetteville
Bachelor of Science in Computer Science | University of Arkansas, Fayetteville

Expected Graduation: May 2027
August 2021 to May 2025

WORK EXPERIENCE

Teaching Assistant, University of Arkansas

August 2025 to Present

- Assisted multiple professors in grading, holding weekly labs, and holding office hours
- Offered tailored assistance on programming assignments, lab exercises, and exam preparation
- Evaluated **20+** students' performance on assignments, maintaining high academic standards and proficiency

Undergraduate Research Assistant, Smart Agricultural & Food Engineering

May 2025 to August 2025

- Operated a Clearpath A200 Husky autonomous mobile robot, integrating ROS-based workflows for sensor fusion, mapping, and navigation as part of an AI-powered environmental monitoring and sanitation system
- Collected and curated multimodal datasets spanning LiDAR, vision, odometry, and IMU data to support downstream perception modeling and sanitation efficacy evaluation
- Generated high-fidelity dynamic facility maps using LiDAR and time-series video analysis, building reproducible and quality-controlled mapping pipelines for environmental monitoring

Undergraduate Research Intern, Washington State University

June 2024 to August 2024

- Fine-tuned YOLOv8 object detection and segmentation models to identify and delineate wheat fields, achieving **82%** detection accuracy and **84%** segmentation accuracy for real-world agricultural deployment
- Processed and analyzed multispectral and RGB imaging data to extract spatial crop features, engineering vegetative indices including the Soil Crop Index and Green Normalized Difference Vegetation Index for crop health assessment
- Applied machine learning techniques to remote sensing data to predict field conditions end-to-end, from raw image ingestion and feature extraction to model evaluation, supporting data-driven precision agriculture

Software Engineer Intern, Arkansas Blue Cross and Blue Shield

June 2023 to August 2023

- Developed Confluent-Kafka consumer client in Python to ingest and process real-time data streams from a pipeline, enabling end-to-end data
- Built Elasticsearch (ELK stack) indexing pipeline in Python to ingest, structure, and export large datasets into multiple file formats, supporting downstream data retrieval and analysis workflows
- Engineered data integration scripts on the Integration Hub team to push and consume structured data across distributed systems, spanning the full pipeline lifecycle from stream ingestion to search indexing, with version-controlled updates via GitHub

PROJECTS

Sports Odd Pipeline

February 2026

- Built a modular ETL pipeline in Python to ingest real-time sports betting odds from major U.S. markets via the Odds API and game results from the ESPN API, storing structured data in a SQLite data warehouse
- Engineered sports market efficiency analytics including close line value, calibration modeling, and best-market frequency tracking, finding pregame odds correctly predicted game outcomes **70%** of the time versus a **50/50** baseline
- Developed an interactive Streamlit dashboard with strategy simulation metrics (ROI, drawdown, profit factor) and one-click ETL refresh, supporting both end-user analytics and admin-level pipeline monitoring

SKILLS

- Programming Languages:** Python, SQL, C++, HTML/CSS
- Machine Learning & Libraries:** PyTorch, Scikit-learn, Pandas, NumPy
- Data Engineering & Visualization:** Kafka, Spark, Elasticsearch, Tableau
- Cloud, DevOps & Tools:** Docker, Azure, Git, GitHub